Chapter 4
The z-Transform

Contents

4.1 Introduction

- The z-transform is the discrete-time counterpart of the Laplace transform.
- It is primary tool for analysis/design in DSP and discrete-time control.
- It is especially useful for solving linear constant-coefficient difference equations (LCCDEs).
- It provides insight into system properties: *stability*, *time-domain response*, *frequency response*, and more.

4.2 The Bilateral z-Transform

Definition (bilateral z-transform)

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}, \qquad z \in \mathbb{C}.$$

- Notation: $x[n] \overset{\mathcal{Z}}{\longleftrightarrow} X(z)$ or $X(z) = \mathcal{Z}\{x[n]\}$.
- The *unilateral* (one-sided) z-transform is defined by

$$X_u(z) = \sum_{n=0}^{\infty} x[n]\, z^{-n}.$$

4.2 The Bilateral z-Transform

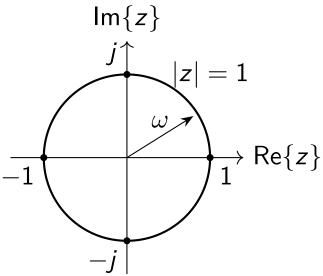
- Write z in polar form: $z = re^{j\omega}$, where $r = |z|$ and $\omega = \arg(z)$.
- Evaluate the z-transform on $z = re^{j\omega}$:

$$X\left(re^{j\omega}\right) = \sum_{n=-\infty}^{\infty} x[n]\,(re^{j\omega})^{-n} = \sum_{n=-\infty}^{\infty} \left(x[n]\,r^{-n}\right) e^{-j\omega n}.$$

- So $X(re^{j\omega})$ is the DTFT of the modified sequence $x[n]\,r^{-n}$.
- Setting $r = 1$ gives the DTFT: $X(e^{j\omega})$ is the z-transform on the *unit circle*.

4.2 The Bilateral z-Transform

- As ω increases by 2π, $z = e^{j\omega}$ traces the unit circle once.
- The mapping $\omega \mapsto e^{j\omega}$ is 2π-periodic.
- Frequency response exists when the ROC contains the unit circle.



4.2.1 Convergence Considerations

Region of Convergence (ROC)

The z-transform converges for those z such that the series is absolutely summable:

$$\sum_{n=-\infty}^{\infty} \left| x[n]\, z^{-n} \right| < \infty.$$

- With $z = re^{j\omega}$, $|z^{-n}| = r^{-n}$ and the convergence depends on $r = |z|$.
- The ROC is typically an annulus: $r_1 < |z| < r_2$ (with r_1 possibly 0 and/or r_2 possibly ∞).
- The DTFT exists if the ROC includes the unit circle ($|z| = 1$).

4.2.1 Convergence Considerations

- In the ROC, $X(z)$ can be represented by a (two-sided) Laurent series in z^{-1}.
- $X(z)$ is analytic (complex differentiable) everywhere in the ROC.
- For many signals/systems, $X(z)$ is a rational function:

$$X(z) = \frac{N(z)}{D(z)}, \qquad N(z), D(z) \text{ polynomials in } z^{-1}.$$

- Poles are the roots of $D(z)$. The ROC *cannot contain poles*.

4.2.1 Convergence Considerations

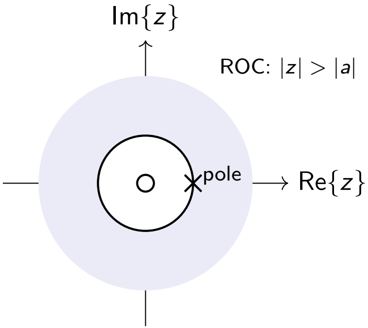
Example 4.1: $x[n] = a^n u[n]$

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}}, \qquad |az^{-1}| < 1.$$

- ROC: $|z| > |a|$.
- Equivalent form: $X(z) = \dfrac{z}{z - a}$ (one pole at $z = a$ and one zero at $z = 0$).

4.2.1 Convergence Considerations

- $x[n] = a^n u[n]$ is right-sided $\Rightarrow$ ROC is outside the outermost pole.
- For $X(z) = \dfrac{z}{z-a}$: pole at $z = a$, zero at $z = 0$.
- ROC: $|z| > |a|$.



Example 4.2: A left-sided exponential

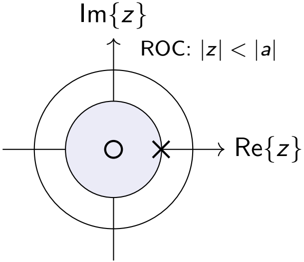
Let $x[n] = -a^n u[-n-1]$ (a time-reversed exponential).

$$X(z) = -\sum_{n=-\infty}^{-1} a^n z^{-n} = -\sum_{m=1}^{\infty} a^{-m} z^m \quad (m=-n) = -\sum_{m=1}^{\infty} \left(\frac{z}{a}\right)^m = \frac{z}{z-a}, \qquad \left|\frac{z}{a}\right| < 1.$$

- ROC: $|z| < |a|$ (inside the pole).
- Important: the *same* rational function $\frac{z}{z-a}$ can correspond to different sequences depending on the ROC.

4.2.1 Convergence Considerations

- $x[n] = -a^n u[-n-1]$ is left-sided $\Rightarrow$ ROC is inside the innermost pole.
- $X(z) = \dfrac{z}{z-a}$ again has a pole at $z = a$ and a zero at $z = 0$.
- ROC: $|z| < |a|$.



Example 4.3: Sum of right-sided exponentials
Let

$$x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(-\frac{1}{3}\right)^n u[n].$$

Using linearity and Example 4.1:

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 + \frac{1}{3}z^{-1}} = \frac{2z(z - \frac{1}{12})}{(z - \frac{1}{2})(z + \frac{1}{3})}.$$

- Individual ROCs: $|z| > \frac{1}{2}$ and $|z| > \frac{1}{3}$.
- Overall ROC is the *intersection*: $|z| > \frac{1}{2}$.

Example 4.4: Mixed left/right-sided terms

Let

$$x[n] = -\left(\frac{1}{2}\right)^n u[-n-1] + \left(-\frac{1}{3}\right)^n u[n].$$

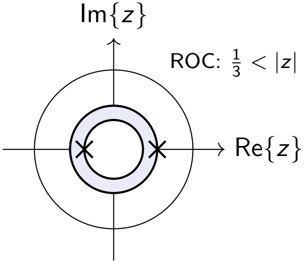
Then (Examples 4.1–4.2):

$$X(z) = \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{1}{1+\frac{1}{3}z^{-1}} = \frac{2z(z-\frac{1}{12})}{(z-\frac{1}{2})(z+\frac{1}{3})}.$$

- ROC for the left-sided term: $|z| < \frac{1}{2}$.
- ROC for the right-sided term: $|z| > \frac{1}{3}$.
- Overall ROC: $\frac{1}{3} < |z| < \frac{1}{2}$.

4.2.1 Convergence Considerations

- Example 4.4 is two-sided $\Rightarrow$ ROC is a ring between poles.
- Poles at $z = \frac{1}{2}$ and $z = -\frac{1}{3}$.
- ROC: $\frac{1}{3} < |z| < \frac{1}{2}$.



4.2.2 Convergence for Finite-Length Sequences

Consider a finite-length sequence supported on $N_1 \leq n \leq N_2$. Then

$$X(z) = \sum_{n=N_1}^{N_2} x[n]\, z^{-n}.$$

- A finite sum converges for all z except possibly $z = 0$ or $z = \infty$ depending on the powers of z.
- If only nonnegative powers of z^{-1} appear (right-sided/causal finite length), then ROC is $|z| > 0$.
- If only negative powers of z^{-1} appear (left-sided finite length), then ROC is $|z| < \infty$.

4.2.2 Convergence for Finite-Length Sequences

Example 4.5: A finite-length exponential

Let

$$x[n] = \begin{cases} a^n, & 0 \le n \le N-1, \\ 0, & \text{otherwise.} \end{cases}$$

Then

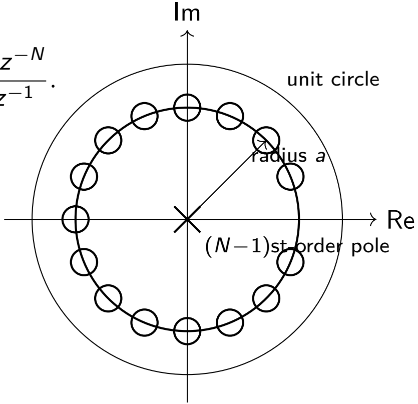
$$X(z) = \sum_{n=0}^{N-1} a^n z^{-n} = \sum_{n=0}^{N-1} (az^{-1})^n = \frac{1-(az^{-1})^N}{1-az^{-1}}.$$

- ROC: $|z| > 0$ (finite-length right-sided sequence).
- The zeros are the Nth roots of unity scaled by a (excluding the pole-zero cancellation when applicable).

4.2.2 Convergence for Finite-Length Sequences

For

$$x[n] = \begin{cases} a^n, & 0 \le n \le N-1, \\ 0, & \text{otherwise}, \end{cases} \qquad X(z) = \frac{1 - a^N z^{-N}}{1 - az^{-1}}.$$



- The roots of $1 - a^N z^{-N}$ lie on a circle of radius a.
- One of those roots cancels the factor $(1 - az^{-1})$.
- The remaining $N-1$ zeros are at

  $$z = a\, e^{j2\pi\ell/N}, \qquad \ell = 1, \ldots, N-1,$$

  together with an $(N-1)$st-order pole at the origin.

4.3 Properties of the ROC

- For rational $X(z)$, the ROC is determined by poles and the time support of $x[n]$.
- The ROC is an annulus centered at the origin: $r_1 < |z| < r_2$.
- The DTFT $X(e^{j\omega})$ exists iff the unit circle ($|z| = 1$) lies in the ROC.

4.3 Properties of the ROC

Property 1: ROC is an annulus centered at the origin

For rational $X(z)$, the ROC has the form $r_1 < |z| < r_2$ (where r_1 can be 0 and/or r_2 can be ∞).

Property 2: ROC contains no poles

Poles are values of z where $X(z) \to \infty$. A convergent power/Laurent series cannot include poles in its ROC.

4.3 Properties of the ROC

Property 3: Finite-duration sequences

If $x[n]$ is finite-duration, then $X(z)$ converges everywhere except possibly at $z = 0$ and/or $z = \infty$:

- Right-sided finite-duration $\Rightarrow$ ROC is all z except possibly $z = 0$ (i.e., $|z| > 0$).
- Left-sided finite-duration $\Rightarrow$ ROC is all z except possibly $z = \infty$ (i.e., $|z| < \infty$).
- Two-sided finite-duration $\Rightarrow$ ROC excludes both $z = 0$ and $z = \infty$ (i.e., $0 < |z| < \infty$).

4.3 Properties of the ROC

Property 4: Right-sided (causal) sequences

If $x[n] = 0$ for $n < n_0$ (right-sided), then the ROC is exterior to the outermost pole:

$$|z| > r_{\max}, \quad r_{\max} = \max_k |p_k|.$$

Property 5: Left-sided sequences

If $x[n] = 0$ for $n > n_0$ (left-sided), then the ROC is interior to the innermost pole:

$$|z| < r_{\min}, \quad r_{\min} = \min_k |p_k|.$$

4.3 Properties of the ROC

Property 6: Two-sided sequences

If $x[n]$ is two-sided, the ROC is between two poles (a ring that excludes poles):

$$r_{\text{min}} < |z| < r_{\text{max}}.$$

Property 7: ROC is connected

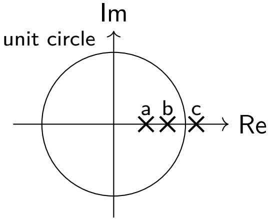
The ROC of $X(z)$ is a connected region (no "holes" other than those created by poles).

4.3 Properties of the ROC

Frequency response and stability

- The DTFT $X(e^{j\omega})$ exists iff the ROC includes the unit circle $|z| = 1$.
- For an LTI system with impulse response $h[n]$ and $H(z) = \mathcal{Z}\{h[n]\}$:
  - BIBO stability $\Leftrightarrow \sum_n |h[n]| < \infty \Leftrightarrow$ ROC includes the unit circle.
  - Causality (right-sided $h[n]$) $\Rightarrow$ ROC is outside the outermost pole (and includes ∞).
  - Stable *and* causal (for rational H) $\Rightarrow$ all poles must lie strictly inside the unit circle.

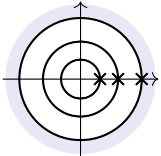
Example 4.6: Possible ROCs (illustration)
Consider a rational $X(z)$ with three poles at radii $|a| < |b| < |c|$ (real axis shown for illustration).



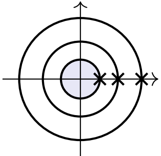
- The ROC must be one of: $|z| > |c|$, $|z| < |a|$, $|a| < |z| < |b|$, or $|b| < |z| < |c|$.

Example 4.6: the four possible ROCs

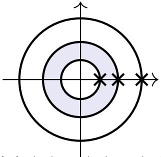
For poles at radii $|a| < |b| < |c|$, the ROC can only be outside the outermost pole, inside the innermost pole, or between adjacent poles.



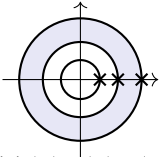
(a) $|z| > |c|$



(b) $|z| < |a|$



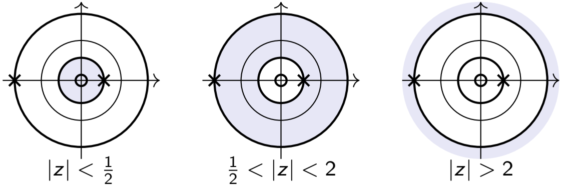
(c) $|a| < |z| < |b|$



(d) $|b| < |z| < |c|$

Example 4.7: stability and causality from the ROC

For the system with poles at $z = \frac{1}{2}$ and $z = -2$, the three valid ROCs are shown below.



- Stability requires the unit circle be in the ROC $\Rightarrow$ only the middle annulus is stable.
- Causality requires the ROC be outside the outermost pole $\Rightarrow$ only $|z| > 2$ is causal.
- This system cannot be both stable and causal.

4.3.1 ROC Summary for Finite-Duration Signals

Signal class	**ROC**	**Comment**
Causal finite-duration	$\lvert z \rvert > 0$	All powers are nonnegative in z^{-1}, so only $z = 0$ is excluded.
Anti-causal finite-duration	$\lvert z \rvert < \infty$	Only negative powers of z^{-1} appear, so only $z = \infty$ is excluded.
Two-sided finite-duration	$0 < \lvert z \rvert < \infty$	Both $z = 0$ and $z = \infty$ are excluded.

For finite-length sequences the ROC contains every finite nonzero point in the z-plane.

4.3.2 ROC Summary for Infinite-Duration Signals

Signal class	ROC	Key fact
Right-sided / causal	$\lvert z\rvert > r_{\text{max}}$	ROC is exterior to the outermost pole.
Left-sided / anti-causal	$\lvert z\rvert < r_{\text{min}}$	ROC is interior to the innermost pole.
Two-sided	$r_1 < \lvert z\rvert < r_2$	ROC lies between poles.

The DTFT exists precisely when the unit circle lies inside the ROC.

4.4 The Inverse z-Transform

- The z-transform is a one-to-one mapping only when the ROC is specified.
- Several practical inversion methods:
  - Inspection (pattern matching using known transform pairs).
  - Partial-fraction expansion (for rational $X(z)$).
  - Power-series expansion (Laurent series).
  - Contour integration.

Inversion integral (bilateral)

$$x[n] = \frac{1}{2\pi j} \oint_{\mathcal{C}} X(z)\, z^{n-1}\, dz,$$

where $\mathcal{C}$ is a closed contour within the ROC encircling the origin once (counterclockwise).

4.4 The Inverse z-Transform

- If $X(z)$ can be decomposed into a sum of known pairs (with compatible ROC), $x[n]$ follows by inspection.
- For rational functions, partial fractions and the ROC tell you which side (right/left) each term corresponds to.

4.4.1 Inspection Method

Common z-transform pairs (with ROC)

$x[n]$	$X(z) = \mathcal{Z}\{x[n]\}$ and ROC
$\delta[n]$	1 (all z)
$u[n]$	$\dfrac{1}{1 - z^{-1}}, \quad \lvert z\rvert > 1$
$-u[-n-1]$	$\dfrac{1}{1 - z^{-1}}, \quad \lvert z\rvert < 1$
$a^n u[n]$	$\dfrac{1}{1 - az^{-1}}, \quad \lvert z\rvert > \lvert a\rvert$
$-a^n u[-n-1]$	$\dfrac{1}{1 - az^{-1}}, \quad \lvert z\rvert < \lvert a\rvert$
$na^n u[n]$	$\dfrac{az^{-1}}{(1 - az^{-1})^2}, \quad \lvert z\rvert > \lvert a\rvert$

Additional transform pairs

$x[n]$	$X(z)$	ROC
$\delta[n-m]$	z^{-m}	all z (except $z=0$ if $m>0$)
$\binom{n+k-1}{k-1} a^n u[n]$	$\dfrac{1}{(1-az^{-1})^k}$	$\|z\| > \|a\|$
$r^n \cos(\omega_0 n) u[n]$	$\dfrac{1 - r\cos\omega_0\, z^{-1}}{1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2}}$	$\|z\| > r$
$r^n \sin(\omega_0 n) u[n]$	$\dfrac{r\sin\omega_0\, z^{-1}}{1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2}}$	$\|z\| > r$

4.4.1 Inspection Method

Additional transform pairs (continued)

$x[n]$	$X(z)$	ROC
$\cosh(bn)u[n]$	$\dfrac{1-\cosh b\, z^{-1}}{1-2\cosh b\, z^{-1}+z^{-2}}$	$\lvert z\rvert > \max\{\lvert e^{b}\rvert, \lvert e^{-b}\rvert\}$
$\sinh(bn)u[n]$	$\dfrac{\sinh b\, z^{-1}}{1-2\cosh b\, z^{-1}+z^{-2}}$	$\lvert z\rvert > \max\{\lvert e^{b}\rvert, \lvert e^{-b}\rvert\}$
$\begin{cases} a^n, & 0 \le n \le N \\ 0, & \text{otherwise} \end{cases}$	$\dfrac{1-a^{N+1}z^{-(N+1)}}{1-az^{-1}}$	$\lvert z\rvert > 0$

4.4.2 Partial Fraction Method

Assume a rational transform

$$X(z) = \frac{N(z)}{D(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}.$$

- Poles are the roots of $D(z)$; zeros are the roots of $N(z)$.
- If $M < N$, then $X(z)$ is a *proper* rational function.
- If $M \geq N$, do long division first and write

$$X(z) = \sum_{r=0}^{M-N} B_r z^{-r} + \frac{N_1(z)}{D(z)},$$

  where $N_1(z)$ has lower order than $D(z)$.
- Polynomial terms correspond to shifted impulses in time.

4.4.2 Partial Fraction Method

Distinct poles

If

$$D(z) = \prod_{k=1}^{N} (1 - d_k z^{-1}),$$

then

$$X(z) = \sum_{k=1}^{N} \frac{A_k}{1 - d_k z^{-1}}, \qquad A_k = (1 - d_k z^{-1}) X(z)\big|_{z=d_k}.$$

Multiple-order poles

If $z = d_i$ is a pole of order s, use

$$\sum_{m=1}^{s} \frac{C_m}{(1 - d_i z^{-1})^m}$$

in the expansion; the coefficients C_m follow from the usual derivative formula for repeated poles.

4.4.2 Partial Fraction Method

General inversion procedure

Given ROC $r_R < |z| < r_L$:

- A polynomial term $B_r z^{-r}$ gives $B_r \, \delta[n - r]$.
- If a pole d_k satisfies $|d_k| \leq r_R$, then

$$\frac{A_k}{1 - d_k z^{-1}} \Longleftrightarrow A_k d_k^n u[n].$$

- If a pole d_k satisfies $|d_k| \geq r_L$, then

$$\frac{A_k}{1 - d_k z^{-1}} \Longleftrightarrow -A_k d_k^n u[-n-1].$$

- Repeated poles are handled the same way, with each term classified as right-sided or left-sided from the ROC.

Example 4.8
Given

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}, \qquad \text{ROC: } |z| > 1.$$

Since numerator and denominator have the same order, perform long division:

$$X(z) = 2 + \frac{5z^{-1} - 1}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}.$$

Factor the denominator:

$$1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2} = \left(1 - \frac{1}{2}z^{-1}\right)\left(1 - z^{-1}\right).$$

4.4.2 Partial Fraction Method

Example 4.8 (continued)
Partial-fraction expansion:

$$X(z) = 2 + \frac{A_1}{1 - \frac{1}{2}z^{-1}} + \frac{A_2}{1 - z^{-1}}.$$

Solve for coefficients:

$$5z^{-1} - 1 = A_1(1 - z^{-1}) + A_2\left(1 - \frac{1}{2}z^{-1}\right),$$

$$A_1 = -9, \qquad A_2 = 8.$$

So

$$X(z) = 2 - \frac{9}{1 - \frac{1}{2}z^{-1}} + \frac{8}{1 - z^{-1}}.$$

With ROC $|z| > 1$ (right-sided terms):

$$x[n] = 2\delta[n] - 9\left(\frac{1}{2}\right)^n u[n] + 8u[n].$$

Example 4.9: Same $X(z)$, different ROCs
For the same rational function in Example 4.8, different valid ROCs give different sequences:

- If ROC $|z| < \frac{1}{2}$ (all left-sided):

$$x[n] = 2\delta[n] + 9\left(\frac{1}{2}\right)^n u[-n-1] - 8\,u[-n-1].$$

- If ROC $\frac{1}{2} < |z| < 1$ (mixed):

$$x[n] = 2\delta[n] - 9\left(\frac{1}{2}\right)^n u[n] - 8\,u[-n-1].$$

Example 4.10: Repeated poles
Given

$$X(z) = \frac{1}{(1+z^{-1})(1-z^{-1})^2}, \qquad \text{ROC: } |z| > 1.$$

Use a partial-fraction form for repeated poles:

$$X(z) = \frac{A_1}{1+z^{-1}} + \frac{A_2}{1-z^{-1}} + \frac{A_3}{(1-z^{-1})^2}.$$

(Solve A_1, A_2, A_3 by equating coefficients or using the residue/cover-up method.)

4.4.2 Partial Fraction Method

Distinct complex-conjugate poles

If $X(z)$ has a conjugate pole pair $d_k = r_k e^{j\beta_k}$ and d_k^* with residues A_k and A_k^*, then

$$X_k(z) = \frac{A_k}{1 - d_k z^{-1}} + \frac{A_k^*}{1 - d_k^* z^{-1}}.$$

For a right-sided sequence (ROC $|z| > r_k$):

$$\begin{aligned} x_k[n] &= A_k d_k^n u[n] + A_k^* (d_k^*)^n u[n] \\ &= 2|A_k|\, r_k^n \cos(\beta_k n + \alpha_k)\, u[n], \quad A_k = |A_k| e^{j\alpha_k}. \end{aligned}$$

Example 4.11: Inverting a damped sinusoid (causal)
Given

$$X(z) = \frac{1 + z^{-1}}{1 - z^{-1} + \frac{1}{2}z^{-2}}, \qquad \text{ROC: } |z| > \frac{1}{\sqrt{2}}.$$

The denominator factors as $(1 - d_1 z^{-1})(1 - d_2 z^{-1})$ with

$$d_{1,2} = \frac{1}{2} \pm j\frac{1}{2} = \frac{1}{\sqrt{2}} e^{\pm j\pi/4}.$$

After partial fractions, the result is a damped sinusoid of the form $2|A|r^n \cos(\beta n + \alpha)u[n]$ with $r = 1/\sqrt{2}$ and $\beta = \pi/4$.

4.4.3 Power Series expansion

- Expand $X(z)$ as a power (Laurent) series in z^{-1} or z whose coefficients reveal $x[n]$.
- Choice of expansion depends on the ROC (e.g., geometric series requires $|az^{-1}| < 1$ or $|z/a| < 1$).

Geometric-series identity

$$\frac{1}{1-w} = \sum_{n=0}^{\infty} w^n, \qquad |w| < 1.$$

4.4.3 Power Series expansion

Example 4.12
If $X(z) = 1 + z^{-1} + z^{-2} + z^{-3}$, then by inspection

$$x[n] = \delta[n] + \delta[n-1] + \delta[n-2] + \delta[n-3].$$

4.4.3 Power Series expansion

Example 4.13

Given $X(z) = \dfrac{1}{1 - az^{-1}}$:

- If ROC $|z| > |a|$, expand in powers of z^{-1}:

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} \iff x[n] = a^n u[n].$$

- If ROC $|z| < |a|$, rewrite and expand in powers of z:

$$X(z) = \frac{-z/a}{1 - z/a} = -\sum_{n=1}^{\infty} a^{-n} z^n \iff x[n] = -a^n u[-n-1].$$

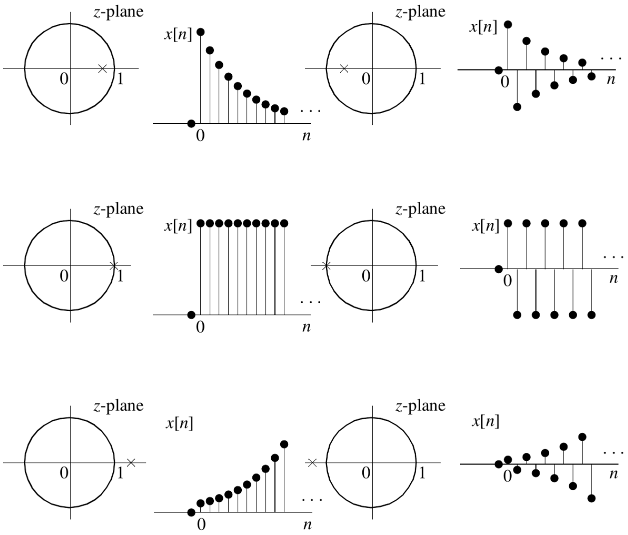
Example 4.14: Logarithm expansion
Use the Taylor series

$$\ln(1+w) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} w^n, \qquad |w| < 1.$$

Let $X(z) = \ln(1 + az^{-1})$. If ROC $|z| > |a|$, then with $w = az^{-1}$:

$$X(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} a^n z^{-n} \iff x[n] = \frac{(-1)^{n-1}}{n} a^n u[n-1].$$

4.4.4 Time-Domain Responses for a Single Real Pole

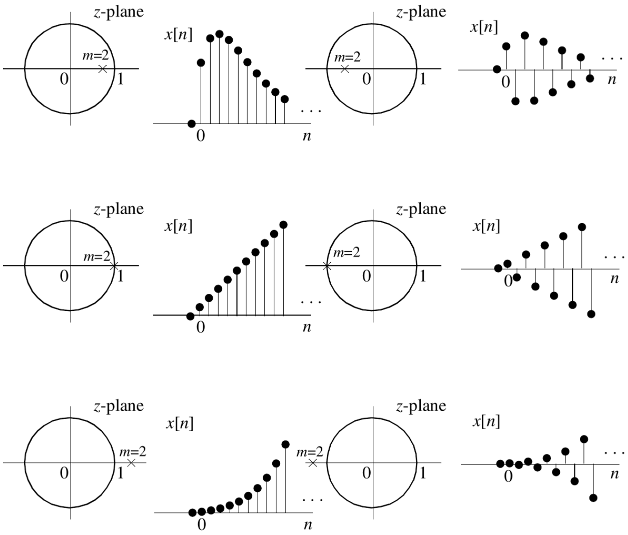


Case 1
$0 < |a| < 1$
ROC $|z| > |a|$:
decaying right-sided
ROC $|z| < |a|$:
growing left-sided

Case 2
$|a| = 1$
constant-magnitude
sequence

Case 3
$1 < |a| < \infty$
ROC $|z| > |a|$:
growing right-sided
ROC $|z| < |a|$:
decaying left-sided

4.4.5 Time-Domain Responses for a Double Real Pole



Repeated pole
time response
$\propto (n+1)a^n$

Case 1
$0 < |a| < 1$
decaying / damped response

Case 2
$|a| = 1$
linear growth in magnitude

Case 3
$1 < |a| < \infty$
rapidly growing response

4.4.6 Conjugate Pole Pair: Transform Pair

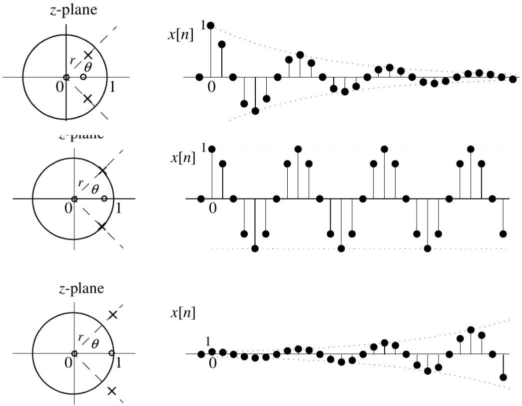
A conjugate pole pair at $z = re^{\pm j\omega_0}$ generates an oscillatory sequence. The radius r sets the exponential growth/decay, while ω_0 sets the oscillation frequency.

$$r^n \cos(\omega_0 n)u[n] \iff \frac{1-(r\cos\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}, \qquad |z| > r.$$

- $0 < r < 1$: damped oscillation.
- $r = 1$: sustained oscillation.
- $r > 1$: growing oscillation.

4.4.6 Time-Domain Responses for a Conjugate Pole Pair

Pole angle determines oscillation frequency: $\theta = \omega_0$



Case 1
$0 < r < 1$
ROC $|z| > r$
damped oscillation

Case 2
$r = 1$
ROC $|z| > 1$
sustained oscillation

Case 3
$r > 1$
ROC $|z| > r$
growing oscillation

4.5 z-Transform Properties

- Many z-transform properties mirror Laplace/DTFT properties, but the ROC must be tracked.
- Below, assume $x[n] \overset{\mathcal{Z}}{\longleftrightarrow} X(z)$ with ROC $\mathcal{R}_x$.

4.5 z-Transform Properties

Linearity

If $x_1[n] \overset{\mathcal{Z}}{\longleftrightarrow} X_1(z)$ and $x_2[n] \overset{\mathcal{Z}}{\longleftrightarrow} X_2(z)$, then

$$ax_1[n] + bx_2[n] \overset{\mathcal{Z}}{\longleftrightarrow} aX_1(z) + bX_2(z).$$

Time shifting

If $x[n] \overset{\mathcal{Z}}{\longleftrightarrow} X(z)$, then

$$x[n-n_0] \overset{\mathcal{Z}}{\longleftrightarrow} z^{-n_0}X(z).$$

- ROC is the same as $\mathcal{R}_x$ except possibly at $z=0$ or $z=\infty$ if shifting introduces/removes powers.

Example 4.16

Find $x[n]$ for

$$X(z) = \frac{z^{-1}}{1 - \frac{1}{4}z^{-1}}, \qquad \text{ROC: } |z| > \frac{1}{4}.$$

Rewrite:

$$X(z) = -4 + \frac{4}{1 - \frac{1}{4}z^{-1}}.$$

Thus

$$x[n] = -4\delta[n] + 4\left(\frac{1}{4}\right)^n u[n] = 4\left(\frac{1}{4}\right)^n u[n-1].$$

Multiplication by an exponential (scaling)

If $x[n] \overset{\mathcal{Z}}{\longleftrightarrow} X(z)$, then for $a \neq 0$:

$$a^n x[n] \overset{\mathcal{Z}}{\longleftrightarrow} X\left(\frac{z}{a}\right).$$

- ROC scales accordingly: if $\mathcal{R}_x$ is $r_1 < |z| < r_2$, then the new ROC is $|a|r_1 < |z| < |a|r_2$.

Differentiation (multiplication by n in time)

If $x[n] \overset{\mathcal{Z}}{\longleftrightarrow} X(z)$, then

$$n\, x[n] \overset{\mathcal{Z}}{\longleftrightarrow} -z\frac{dX(z)}{dz}.$$

4.5 z-Transform Properties

Example 4.17: $x[n] = na^n u[n]$

Since $a^n u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \dfrac{1}{1 - az^{-1}}$ (ROC $|z| > |a|$),

$$
\begin{aligned}
X(z) = \mathcal{Z}\{na^n u[n]\} &= -z\frac{d}{dz}\left(\frac{1}{1 - az^{-1}}\right) \\
&= \frac{az^{-1}}{(1 - az^{-1})^2} = \frac{az}{(z - a)^2}, \qquad |z| > |a|.
\end{aligned}
$$

4.5 z-Transform Properties

Conjugation

If $x[n] \stackrel{\mathcal{Z}}{\longleftrightarrow} X(z)$, then

$$x^*[n] \stackrel{\mathcal{Z}}{\longleftrightarrow} X^*(z^*),$$

with the same ROC (reflected appropriately in the z-plane).

Time reversal

If $x[n] \overset{\mathcal{Z}}{\longleftrightarrow} X(z)$, then

$$x[-n] \overset{\mathcal{Z}}{\longleftrightarrow} X(z^{-1}).$$

- If $\mathcal{R}_x$ is $r_1 < |z| < r_2$, the reversed ROC is $\frac{1}{r_2} < |z| < \frac{1}{r_1}$.

Example 4.18: $x[n] = u[-n]$

Recall $u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \dfrac{1}{1 - z^{-1}}$ with ROC $|z| > 1$. Using time reversal:

$$u[-n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1 - z} = -\frac{z^{-1}}{1 - z^{-1}}, \qquad \text{ROC } |z| < 1.$$

4.5 z-Transform Properties

Convolution in time $\leftrightarrow$ multiplication in z

If $y[n] = x_1[n] * x_2[n] = \sum_{k=-\infty}^{\infty} x_1[k]\, x_2[n-k]$, then

$$Y(z) = X_1(z)\, X_2(z).$$

- ROC includes at least the intersection $\mathcal{R}_{x_1} \cap \mathcal{R}_{x_2}$, possibly larger if pole-zero cancellations occur.

4.5 z-Transform Properties

Example 4.19

Let $x_1[n] = a^n u[n]$ and $x_2[n] = u[n]$. Then

$$X_1(z) = \frac{1}{1 - az^{-1}}, \quad |z| > |a|,$$

$$X_2(z) = \frac{1}{1 - z^{-1}}, \quad |z| > 1,$$

$$Y(z) = X_1(z)X_2(z) = \frac{1}{(1 - az^{-1})(1 - z^{-1})}.$$

With ROC $|z| > \max(1, |a|)$ (right-sided), partial fractions give

$$y[n] = \frac{1 - a^{n+1}}{1 - a}\, u[n].$$

4.6 Inverse z-Transform Using Contour Integration

- The inversion integral

$$x[n] = \frac{1}{2\pi j} \oint_{\mathcal{C}} X(z)\, z^{n-1}\, dz$$

  can be evaluated using tools from complex analysis.
- Key idea: choose a contour $\mathcal{C}$ inside the ROC and apply Cauchy/residue theorems.

4.6.1 Complex Variable Background

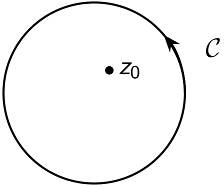
Neighborhood and analyticity

- A *neighborhood* of z_0 is a set of points with $|z - z_0| < \epsilon$ for some $\epsilon > 0$.
- A function $F(z)$ is *analytic* at z_0 if it is complex differentiable in a neighborhood of z_0.
- A point where $F(z)$ is not analytic is a *singular point* (e.g., a pole).

Cauchy's Integral Theorem

If $F(z)$ is analytic on and inside a simple closed contour $\mathcal{C}$, then

$$\oint_{\mathcal{C}} F(z)\, dz = 0.$$



4.6.2 Cauchy's Integral Theorem

Cauchy integral formula (special case)

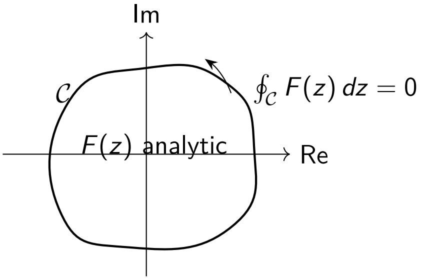
If $F(z)$ is analytic on and inside $\mathcal{C}$ and z_0 is inside $\mathcal{C}$, then

$$F(z_0) = \frac{1}{2\pi j} \oint_{\mathcal{C}} \frac{F(z)}{z - z_0}\, dz.$$

- If the integrand has a singularity inside the contour, the integral need not be zero.

4.6.2 Cauchy's Integral Theorem

- If $F(z)$ is analytic on and inside $\mathcal{C}$, then
$$\oint_{\mathcal{C}} F(z)\, dz = 0.$$
- If an isolated singularity z_0 lies inside $\mathcal{C}$, the integral need not vanish.

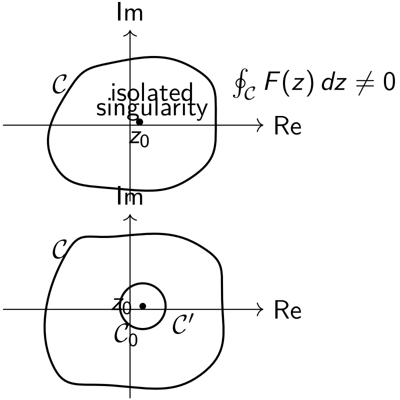


4.6.2 Cauchy's Integral Theorem

- With an isolated singularity inside $\mathcal{C}$,

$$\oint_{\mathcal{C}} F(z)\, dz \neq 0.$$

- We can deform the contour to avoid z_0 by introducing a small circle $\mathcal{C}_0$ around the singularity.
- Then the sum of the contour integrals on the deformed path is zero.



Residue theorem (isolated poles)

If $F(z)$ is analytic on and inside $\mathcal{C}$ except for isolated poles $\{p_k\}$ inside $\mathcal{C}$, then

$$\oint_{\mathcal{C}} F(z)\,dz = 2\pi j \sum_k \operatorname*{Res}_{z=p_k}\{F(z)\}.$$

Residue formulas

- Simple pole at $z = p$: $\operatorname*{Res}_{z=p}\{F(z)\} = \lim_{z\to p}(z-p)F(z)$.
- Pole of order s at $z = p$: $\operatorname*{Res}_{z=p}\{F(z)\} = \dfrac{1}{(s-1)!}\left[\dfrac{d^{\,s-1}}{dz^{\,s-1}}(z-p)^s F(z)\right]_{z=p}$.

4.6.4 Complex Inversion Integral

- In the ROC, $X(z)$ has a Laurent expansion:

$$X(z) = \sum_{k=-\infty}^{\infty} x[k]\, z^{-k}.$$

- Multiply by z^{n-1} and integrate around $\mathcal{C}$ in the ROC:

$$\oint_{\mathcal{C}} X(z)\, z^{n-1}\, dz = \sum_{k=-\infty}^{\infty} x[k] \oint_{\mathcal{C}} z^{n-k-1}\, dz.$$

- The contour integral is

$$\frac{1}{2\pi j} \oint_{\mathcal{C}} z^{n-k-1}\, dz = \begin{cases} 1, & k = n, \\ 0, & k \neq n, \end{cases}$$

which selects $x[n]$.

4.6.4 Complex Inversion Integral

Complex inversion integral (restated)

$$x[n] = \frac{1}{2\pi j} \oint_{\mathcal{C}} X(z)\, z^{n-1}\, dz.$$

- If the ROC includes the unit circle, choose $\mathcal{C} : z = e^{j\omega}$ to obtain the inverse DTFT:

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n}\, d\omega.$$

4.6.5 Evaluating the Inversion Integral

- Let $F_n(z) = X(z)\, z^{n-1}$. Then

$$x[n] = \frac{1}{2\pi j} \oint_{\mathcal{C}} F_n(z)\, dz.$$

- For rational $X(z)$, $F_n(z)$ is rational and its poles are the poles of $X(z)$ (and possibly $z = 0$ if $n \leq 0$).
- Choose $\mathcal{C}$ within the ROC and sum the residues of $F_n(z)$ for the poles inside $\mathcal{C}$.

Example 4.20: $X(z) = \dfrac{1}{1 - az^{-1}}, \; |z| > |a|$

Write $X(z) = \dfrac{z}{z - a}$. Then

$$x[n] = \frac{1}{2\pi j} \oint_{\mathcal{C}} \frac{z^n}{z - a}\, dz.$$

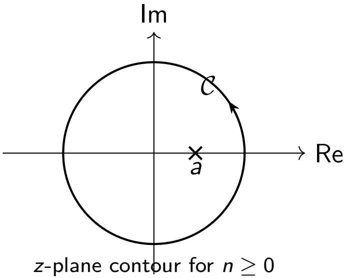
Example 4.20: $n \geq 0$
For

$$X(z) = \frac{1}{1 - az^{-1}}, \qquad |z| > |a|,$$

the inversion integral is

$$x[n] = \frac{1}{2\pi j} \oint_{\mathcal{C}} \frac{z^n}{z - a} \, dz.$$

When $n \geq 0$, the contour encloses the simple pole at $z = a$, so the residue is a^n.



z-plane contour for $n \geq 0$

Example 4.20 (for $n < 0$)
For negative indices it is convenient to change variables to $z = 1/p$ in the inversion integral. Then

$$X(1/p) = \frac{1}{1 - ap}, \qquad |p| < |1/a|,$$

and

$$x[n] = \frac{1}{2\pi j} \oint_{\mathcal{C}'} X(1/p)\, p^{n-1}\, dp.$$

When $n < 0$, the transformed integrand has no poles inside $\mathcal{C}'$, so

$$x[n] = 0, \qquad n < 0.$$

Together with the $n \geq 0$ case, this gives $x[n] = a^n u[n]$.

4.6.5 Evaluating the Inversion Integral

Example 4.20 (for $n < 0$)
It is convenient to change variables to $z = 1/p$ in the inversion integral. Then

$$X(1/p) = \frac{1}{1 - ap}, \qquad |p| < |1/a|,$$

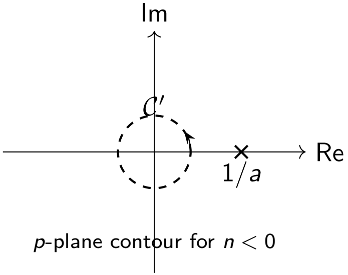
and

$$x[n] = \frac{1}{2\pi j} \oint_{\mathcal{C}'} X(1/p)\, p^{n-1}\, dp.$$

When $n < 0$, the transformed integrand has no poles inside $\mathcal{C}'$, so

$$x[n] = 0, \qquad n < 0.$$

Then, $x[n] = a^n u[n]$.



p-plane contour for $n < 0$

Example 4.21 (setup)
Given

$$X(z) = \frac{z^3 + 2z^2 + 3z}{(z - \frac{1}{2})(z^2 - 2z + 4)}, \qquad \text{ROC: } |z| > 2.$$

The inversion integral is

$$x[n] = \frac{1}{2\pi j} \oint_{\mathcal{C}} z^n \frac{z^2 + 2z + 3}{(z - \frac{1}{2})(z^2 - 2z + 4)}\, dz.$$

Choose $\mathcal{C}$ as a circle of radius $r > 2$ (within the ROC), enclosing the poles at $z = \frac{1}{2}$ and $z = 1 \pm j\sqrt{3}$.

Example 4.21 (residues for $n \geq 0$)
Poles: $p_1 = \frac{1}{2}$, $p_{2,3} = 1 \pm j\sqrt{3}$ (with $|p_{2,3}| = 2$). For $n \geq 0$ and ROC $|z| > 2$,

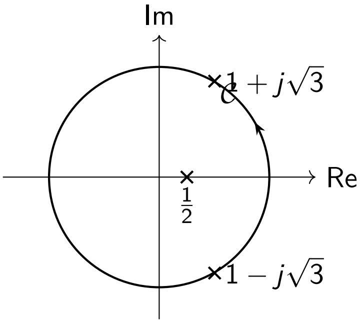
$$x[n] = \sum_{k=1}^{3} \operatorname*{Res}_{z=p_k} \left\{ z^n \frac{z^2 + 2z + 3}{(z - \frac{1}{2})(z^2 - 2z + 4)} \right\}.$$

The residues evaluate to

$$R_1 = \frac{17}{13} \left(\frac{1}{2}\right)^n,$$
$$R_2 = \frac{-1 - j9\sqrt{3}}{13} (1 + j\sqrt{3})^n,$$
$$R_3 = \frac{-1 + j9\sqrt{3}}{13} (1 - j\sqrt{3})^n.$$

So, for $n \geq 0$: $x[n] = (R_1 + R_2 + R_3)u[n]$.

4.6.5 Evaluating the Inversion Integral



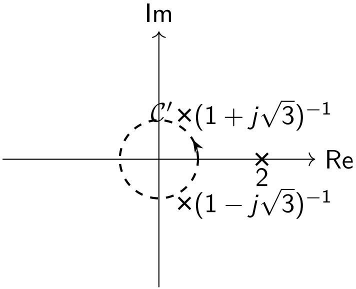
4.6.5 Evaluating the Inversion Integral

Example 4.21: transformed contour for $n < 0$

After $z \mapsto 1/p$, the contour becomes a small circle in the p-plane. The corresponding poles are

$$p_1 = 2, \qquad p_{2,3} = (1 \pm j\sqrt{3})^{-1}.$$

For the ROC $|z| > 2$, none of these poles lie inside $\mathcal{C}'$, hence $x[n] = 0$ for $n < 0$.



Example 4.21 (real form)
Combine the conjugate pair into a real damped sinusoid ($|1 \pm j\sqrt{3}| = 2$):

$$x[n] = \left[\frac{17}{13}\left(\frac{1}{2}\right)^n + 2.418\,(2)^n \cos\left(\frac{\pi}{3}n - 97.3^\circ\right)\right] u[n].$$

(Because ROC is $|z| > 2$, the sequence is right-sided and $x[n] = 0$ for $n < 0$.)